

A Hand-Checkable Proof of Positivity for the Aggregate Curvature Floor

Abstract

We prove, without numerical computation, the strict positivity of the one-variable function that supplies the curvature floor in the aggregate-tail argument for spherical shells. Every estimate is reduced to an exact rational comparison. In fact, on the full required interval we obtain the uniform lower bound

$$F(\rho) > \frac{1549}{84150} > 0.$$

1 Definitions and elementary estimates

For $0 \leq s \leq 1$, define

$$G_1(s) := \frac{1}{\pi} \left(\sqrt{1-s^2} - s \arccos s \right). \quad (1)$$

For $0 < \rho < 1$, put

$$\eta_\rho := G_1(\max\{\rho, 1/2\}), \quad d_\rho := \frac{2 \arcsin \rho}{\pi}, \quad b_\rho := \frac{2\pi\rho}{1-\rho}. \quad (2)$$

The function to be estimated is

$$F(\rho) := 2\rho\eta_\rho + d_\rho\rho^2 - \frac{3(1+d_\rho)b_\rho}{800}. \quad (3)$$

The interval in question is well inside $(0, 1)$, since

$$0 < \frac{7}{51} < \frac{1}{2} < \frac{39}{50} < 1;$$

the two nontrivial middle comparisons are respectively $14 < 51$ and $25 < 39$. Thus all quantities in (2) and (3) are well defined there.

We first record all numerical inequalities used below, together with exact proofs.

Lemma 1 (Rational bounds for π and square roots). *One has*

$$\pi < \frac{22}{7}, \quad \sqrt{2} > \frac{7}{5}, \quad \sqrt{3} > \frac{5}{3}, \quad \sqrt{\frac{11}{50}} > \frac{7}{15}. \quad (4)$$

Proof. Polynomial division gives the identity

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}.$$

The left-hand side is positive for $0 < x < 1$. Hence

$$\begin{aligned} 0 &< \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx \\ &= \frac{1}{7} - \frac{4}{6} + 1 - \frac{4}{3} + 4 - 4 \arctan(1) \\ &= \frac{22}{7} - \pi, \end{aligned}$$

where $\arctan(1) = \pi/4$. This proves $\pi < 22/7$.

All three square-root estimates follow by squaring positive quantities:

$$2 - \left(\frac{7}{5}\right)^2 = \frac{1}{25} > 0, \quad 3 - \left(\frac{5}{3}\right)^2 = \frac{2}{9} > 0,$$

and

$$\frac{11}{50} - \left(\frac{7}{15}\right)^2 = \frac{11}{50} - \frac{49}{225} = \frac{1}{450} > 0.$$

□

Lemma 2 (Integral representation and a lower bound for G_1). *For $0 \leq s \leq 1$,*

$$G_1(s) = \frac{1}{\pi} \int_s^1 \arccos t \, dt \geq \frac{2\sqrt{2}}{3\pi} (1-s)^{3/2}. \quad (5)$$

In particular, if $1/2 \leq s \leq 39/50$, then

$$G_1(s) > \frac{343}{11250}. \quad (6)$$

Proof. The antiderivative

$$\int \arccos t \, dt = t \arccos t - \sqrt{1-t^2}$$

shows directly that

$$\int_s^1 \arccos t \, dt = \sqrt{1-s^2} - s \arccos s,$$

which proves the first equality in (5).

For $0 \leq t \leq 1$, let $y = \arccos t$, so $0 \leq y \leq \pi/2$. Since $\sin u \leq u$ for $u \geq 0$,

$$\sqrt{2(1-t)} = \sqrt{2(1-\cos y)} = 2 \sin \frac{y}{2} \leq y = \arccos t.$$

Integrating this pointwise inequality gives

$$\begin{aligned} G_1(s) &\geq \frac{1}{\pi} \int_s^1 \sqrt{2(1-t)} \, dt \\ &= \frac{2\sqrt{2}}{3\pi} (1-s)^{3/2}, \end{aligned}$$

which completes (5). For completeness, $\sin u \leq u$ follows because the derivative of $u - \sin u$ is $1 - \cos u \geq 0$, and the function vanishes at zero.

Now assume $1/2 \leq s \leq 39/50$. Then $1 - s \geq 11/50$. Lemma 1 yields

$$\begin{aligned}
G_1(s) &\geq \frac{2\sqrt{2}}{3\pi}(1-s)^{3/2} \\
&\geq \frac{2\sqrt{2}}{3\pi} \left(\frac{11}{50}\right)^{3/2} \\
&> \frac{2(7/5)}{3(22/7)} \frac{11}{50} \frac{7}{15} \\
&= \frac{49}{165} \frac{77}{750} = \frac{343}{11250}.
\end{aligned}$$

Here the last equality follows by cancelling a factor 11 from 3773/123750. $\square$

2 Strict positivity of the curvature floor

Theorem 3. *For every*

$$\frac{7}{51} \leq \rho \leq \frac{39}{50},$$

the function in (3) satisfies

$$F(\rho) > \frac{1549}{84150} > 0. \quad (7)$$

Proof. Because $\rho > 0$, substitution of $b_\rho = 2\pi\rho/(1-\rho)$ into (3) and division by ρ give the exact identity

$$\frac{F(\rho)}{\rho} = 2\eta_\rho + d_\rho\rho - \frac{3\pi(1+d_\rho)}{400(1-\rho)}. \quad (8)$$

We treat the two branches in the definition of η_ρ separately.

First branch: $7/51 \leq \rho \leq 1/2$. Here $\max\{\rho, 1/2\} = 1/2$, and therefore

$$\begin{aligned}
\eta_\rho &= G_1(1/2) = \frac{1}{\pi} \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \arccos \frac{1}{2} \right) \\
&= \frac{\sqrt{3}}{2\pi} - \frac{1}{6},
\end{aligned} \quad (9)$$

because $\arccos(1/2) = \pi/3$. Since $\arcsin$ is increasing on $[0, 1]$,

$$0 < d_\rho = \frac{2 \arcsin \rho}{\pi} \leq \frac{2 \arcsin(1/2)}{\pi} = \frac{1}{3}.$$

Also $1 - \rho \geq 1/2$, so $(1 - \rho)^{-1} \leq 2$. It follows that

$$\frac{3\pi(1+d_\rho)}{400(1-\rho)} \leq \frac{3\pi(4/3)2}{400} = \frac{\pi}{50} < \frac{11}{175}. \quad (10)$$

On the other hand, Lemma 1 gives

$$2\eta_\rho = \frac{\sqrt{3}}{\pi} - \frac{1}{3} > \frac{5/3}{22/7} - \frac{1}{3} = \frac{35}{66} - \frac{22}{66} = \frac{13}{66}. \quad (11)$$

The middle term $d_\rho \rho$ in (8) is nonnegative. Combining (8), (10), and (11), we obtain

$$\frac{F(\rho)}{\rho} > \frac{13}{66} - \frac{11}{175} = \frac{2275 - 726}{11550} = \frac{1549}{11550}. \quad (12)$$

Since $\rho \geq 7/51$ and $11550 = 7 \cdot 1650$, this implies

$$F(\rho) > \frac{7}{51} \frac{1549}{11550} = \frac{1549}{84150}. \quad (13)$$

Second branch: $1/2 \leq \rho \leq 39/50$. Here $\max\{\rho, 1/2\} = \rho$, so $\eta_\rho = G_1(\rho)$. Lemma 2 gives

$$2\eta_\rho > \frac{343}{5625}. \quad (14)$$

Monotonicity of arcsin, together with $\rho \geq 1/2$, gives

$$d_\rho \geq \frac{1}{3}, \quad d_\rho \rho \geq \frac{1}{6}. \quad (15)$$

For the required upper bound on d_ρ , observe that

$$\left(\frac{39}{50}\right)^2 = \frac{1521}{2500} < \frac{1875}{2500} = \frac{3}{4} = \left(\frac{\sqrt{3}}{2}\right)^2.$$

Both sides are positive, hence $39/50 < \sqrt{3}/2$. Consequently

$$d_\rho \leq \frac{2 \arcsin(39/50)}{\pi} < \frac{2 \arcsin(\sqrt{3}/2)}{\pi} = \frac{2}{3}. \quad (16)$$

Finally, $1 - \rho \geq 11/50$, so $(1 - \rho)^{-1} \leq 50/11$. From (16) and Lemma 1,

$$\begin{aligned} \frac{3\pi(1 + d_\rho)}{400(1 - \rho)} &< \frac{3\pi(5/3)}{400} \frac{50}{11} = \frac{5\pi}{88} \\ &< \frac{5(22/7)}{88} = \frac{5}{28}. \end{aligned} \quad (17)$$

Substitution of (14), (15), and (17) into (8) yields

$$\begin{aligned} \frac{F(\rho)}{\rho} &> \frac{343}{5625} + \frac{1}{6} - \frac{5}{28} \\ &= \frac{9604 + 26250 - 28125}{157500} \\ &= \frac{7729}{157500}. \end{aligned} \quad (18)$$

Since $\rho \geq 1/2$,

$$F(\rho) > \frac{7729}{315000}. \quad (19)$$

The high-branch lower bound is stronger than the one asserted in (7), because exact subtraction gives

$$\frac{7729}{315000} - \frac{1549}{84150} = \frac{120341}{19635000} > 0.$$

The two branches cover the full interval. Their definitions agree at $\rho = 1/2$, because

$$G_1(\max\{1/2, 1/2\}) = G_1(1/2).$$

Consequently, (13) and (19) prove (7) everywhere on $[7/51, 39/50]$. $\square$